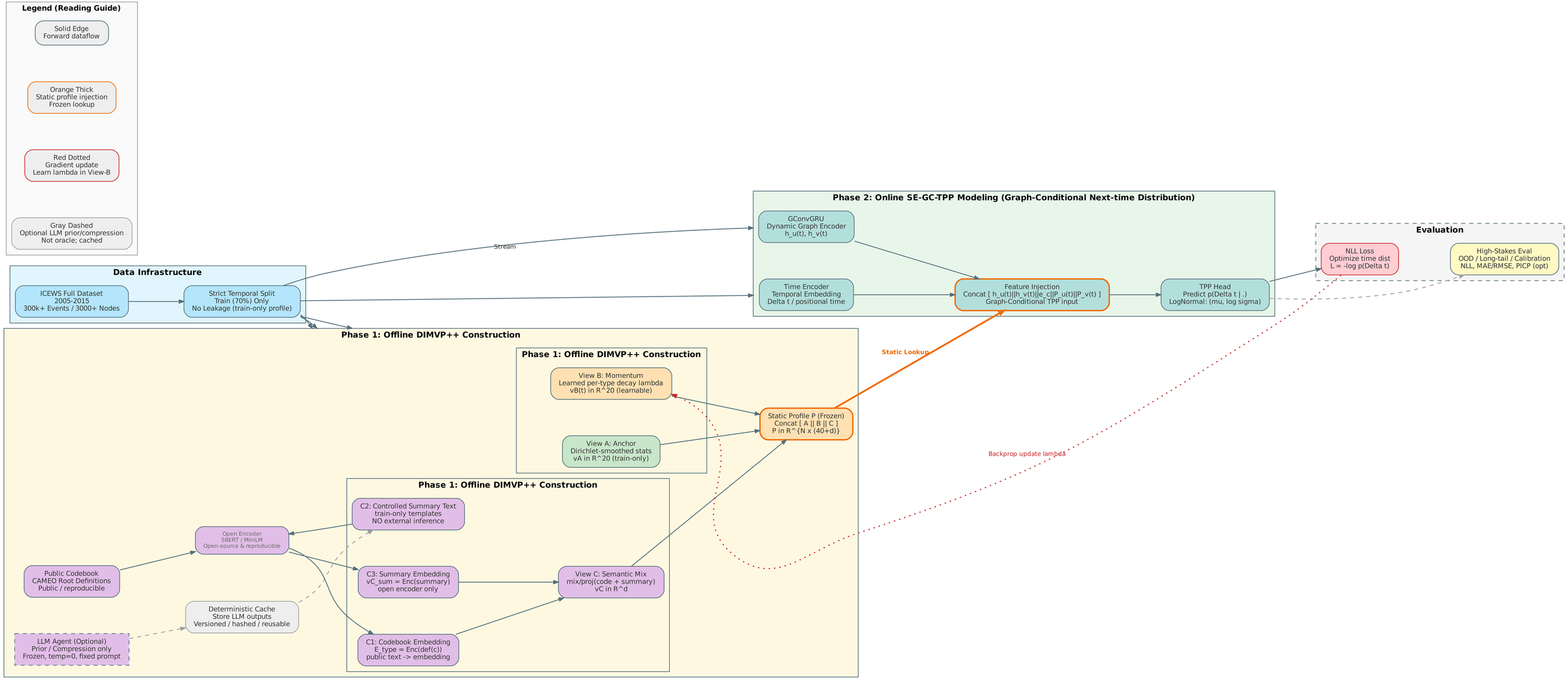
Legend (Reading Guide)
Solid Edge
Forward dataflow
Orange Thick
Static profile injection
Frozen lookup
Red Dotted
Gradient update
Learn lambda in View-B
Gray Dashed
Optional LLM prior/compression
Not oracle; cached
Data Infrastructure
ICEWS Full Dataset
2005-2015
300k+ Events / 3000+ Nodes
Strict Temporal Split
Train (70%) Only
No Leakage (train-only profile)
Stream
Phase 2: Online SE-GC-TPP Modeling (Graph-Conditional Next-time Distribution)
GConvGRU
Dynamic Graph Encoder
h_u(t), h_v(t)
Time Encoder
Temporal Embedding
Delta t / positional time
Feature Injection
Concat [h_u(t)||h_v(t)||e_c||P_u(t)||P_v(t)]
Graph-Conditional TPP input
TPP Head
Predict p(Delta t | .)
LogNormal: (mu, log sigma)
Evaluation
NLL Loss
Optimize time dist
L = -log p(Delta t)
High-Stakes Eval
OOD / Long-tail / Calibration
NLL, MAE/RMSE, PICP (opt)
Phase 1: Offline DIMVP++ Construction
Static Lookup
Phase 1: Offline DIMVP++ Construction
View B: Momentum
Learned per-type decay lambda
vB(t) in R^20 (learnable)
View A: Anchor
Dirichlet-smoothed stats
vA in R^20 (train-only)
Static Profile P (Frozen)
Concat [A || B || C]
P in R^{N x (40+d)}
Backprop update lambda
Phase 1: Offline DIMVP++ Construction
C2: Controlled Summary Text
train-only templates
NO external inference
Open Encoder
SBERT / MiniLM
Open-source & reproducible
Public Codebook
CAMEO Root Definitions
Public / reproducible
C3: Summary Embedding
vC_sum = Enc(summary)
open encoder only
View C: Semantic Mix
mix/proj(code + summary)
vC in R^d
Deterministic Cache
Store LLM outputs
Versioned / hashed / reusable
LLM Agent (Optional)
Prior / Compression only
Frozen, temp=0, fixed prompt
C1: Codebook Embedding
E_type = Enc(def(c))
public text -> embedding